



Linux

Basics (04)

Permissions and

Files

TYPES OF FILES:

The file type can be determined from the first letter/character in Notations when you type 'll' and have files . Such as

-rw-r--r-- ('-'represent regular file)

1.REGULAR FILE (-):

Contains data, text, or program code.

Exp:

file.txt (text file), script.sh (shell script)

Notation Example:

-rw-r--r--

2. BLOCK DEVICE (B):

A block device file represents a device that performs data I/O operations in blocks.

Exp:

hard drives, USB drives

Notation Example:

brw-rw-- --

3. CHARACTER DEVICE (C):

A character device file represents a device that performs data I/O operations one character at a time.

Exp:

keyboards, serial ports

Notation Example:

crw-rw-- --

4.DIRECTORY (D):

A directory is a special file that contains other files and directories. It's used for organizing files.

Notation Example:

drwxr-xr-x

5. SYMBOLIC LINK (L):

A symbolic link is a file that points to another file or directory. It serves as a shortcut or reference.

Notation Example:

lrwxrwxrwx

PERMISSIONS:

Linux file permissions can be represented using numerical values for ease of setting permissions.

UNDERSTANDING NUMERICAL PERMISSIONS

1.Read Permission (4):

Grants the ability to read the contents of the file or directory.

Notation:

r

Representation:

'4' means read permission is set.

2. Write Permission (2):

Grants the ability to modify or delete the file or directory.

Notation:

w

Representation:

‘2’ means write permission is set.

3. Execute Permission (1):

Grants the ability to execute the file (if it is an executable) or access the directory (if it is a directory).

Notation:

x

Representation:

‘1’ means execute permission is set.

COMBINING PERMISSIONS

Permissions are set by adding these values together:

Read and Write (4 + 2 = 6): Grants both read and write permissions.

Notation:

rw-.

Read and Execute (4 + 1 = 5): Grants both read and execute permissions.

Notation:

r-x.

Write and Execute (2 + 1 = 3): Grants both write and execute permissions.

Notation:

WX-.

Read, Write, and Execute (4 + 2 + 1 = 7): Grants all three permissions.

Notation:

rwX.

PERMISSION NOTATION IN OCTAL FORMAT

In octal notation, file permissions are represented by three digits:

First Digit: Permissions for the file owner.

Second Digit: Permissions for the group associated with the file.

Third Digit: Permissions for others (everyone else).

Example:

755 translates to:

Owner: 7 (read, write, and execute: rwx)

Group: 5 (read and execute: r-x)

Others: 5 (read and execute: r-x)

Example Command:

To set permissions using octal notation, you use the `chmod` command. For example, to set permissions to 755:

➤ **chmod 755 [filename]**

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